

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 5

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I N.Sai Sreenu (192365001) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Sai Sreenu

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-I

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

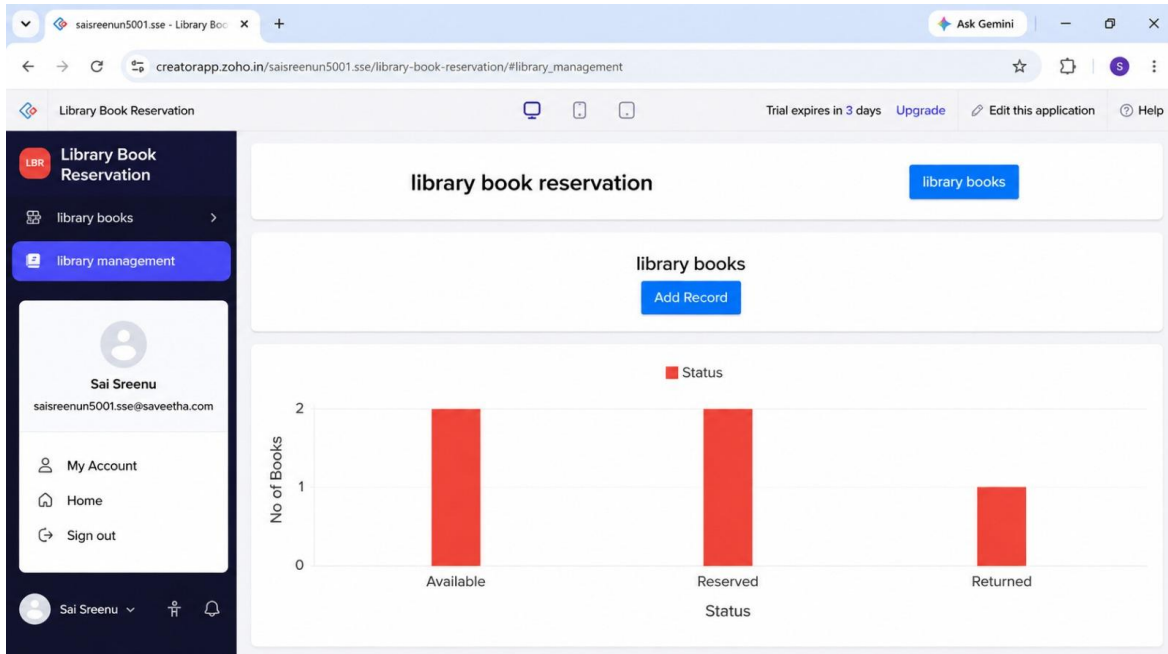
Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Library book reservation system

Dashboard



Form

The form contains the following fields:

- Title of the Book
- Author
- Publication
- Year
- Copies
- Book Category
- Status: ☐ Available, ☐ Reserved, ☐ Returned
- Reservation Date

Report

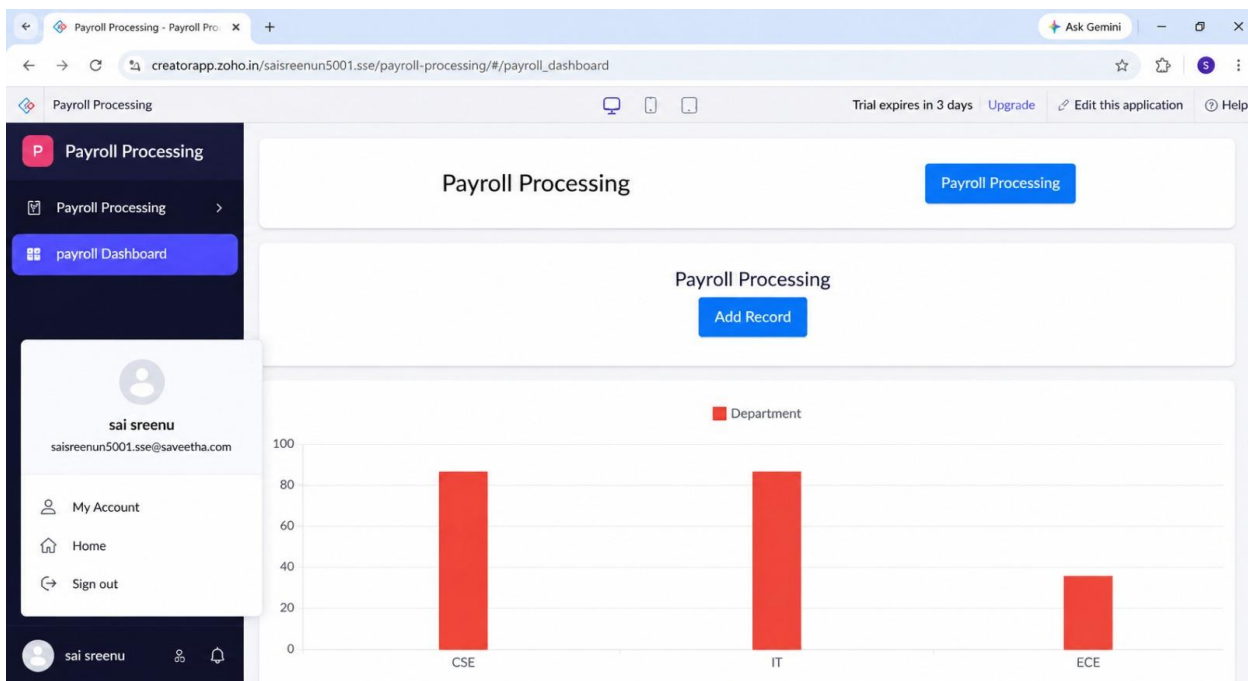
The screenshot shows a web application titled "Library Book Reservation" with a sidebar menu containing "library books" and "library books Report". The main content area displays a table titled "library books Report" with the following data:

Book ID	Title of the Book	Author	Publication	year	Copies	Book Category
5	Gravity	Thareja	Oxford	2022	10	Physics
4	Neural chemistry	Buyya	Oxford	2023	7	Chemistry
3	Python	Reena	Oxford	2019	15	Computer Science
2	DBMS	Korth	Pearson	2022	8	Computer Science
1	cloud computing	Rajkumar	McGraw Hill	2021	10	Computer Science

The sidebar also shows the user profile for Sai Sreenu (saisreenun5001.sse@saveetha.com) and navigation links for "My Account", "Home", and "Sign out". The bottom of the sidebar shows the user's name "Sai Sreenu" and a notification icon.

Experiment 2: Payroll Processing System

Dashboard



Form

Payroll Processing

Payroll Processing

Payroll Processing

Payroll Processing Report

sai sreenu

saisreenun5001.sse@saveetha.com

My Account

Home

Sign out

sai sreenu

Payroll Processing

Employee Name

Department

Basic Pay

HRA

DA

TA

Submit

Report

Payroll Processing

Payroll Processing

Payroll Processing

Payroll Processing Report

sai sreenu

saisreenun5001.sse@saveetha.com

My Account

Home

Sign out

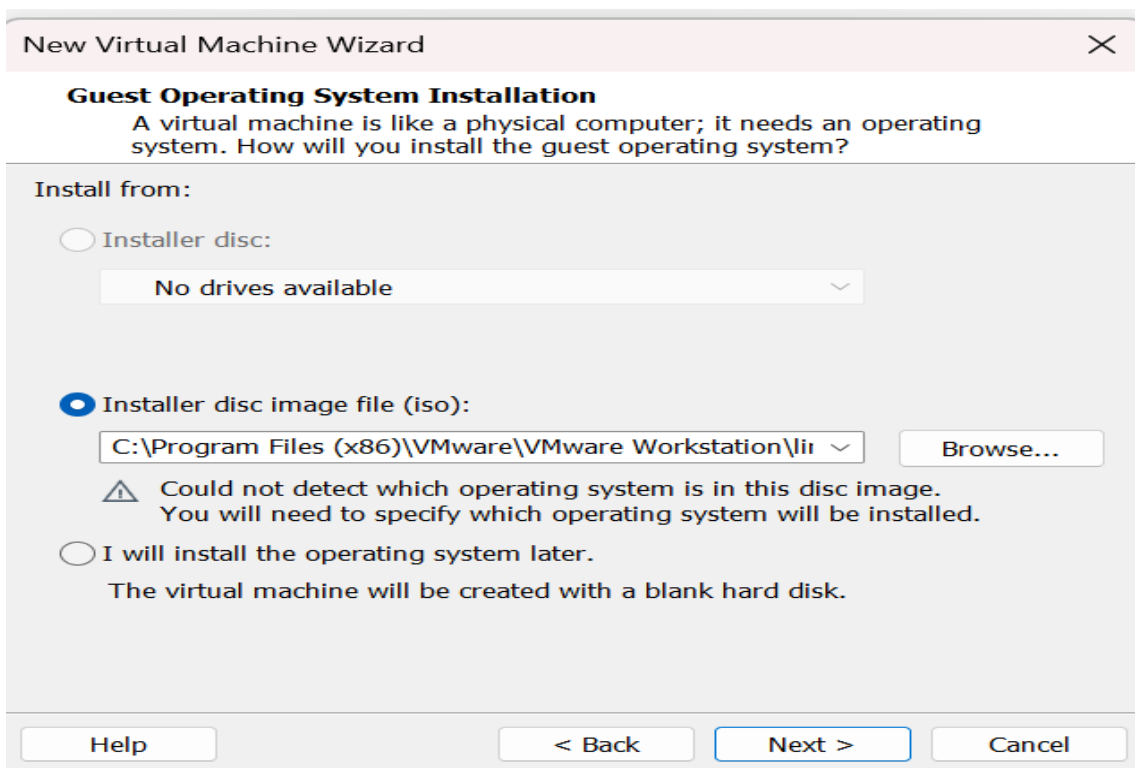
sai sreenu

Payroll Processing Report

Employee ID	Employee Name	Department	Basic Pay	HRA	DA
5	ramu	IT	₹ 20,000.00	₹ 15,000.00	₹ 34,000.00
4	hari	ECE	₹ 20,000.00	₹ 10,000.00	₹ 3,000.00
3	sujith	CSE	₹ 15,000.00	₹ 2,000.00	₹ 3,000.00
2	sumanth	IT	₹ 20,000.00	₹ 15,000.00	₹ 34,000.00
1	karthik	CSE	₹ 30,000.00	₹ 10,000.00	₹ 3,000.00

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Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor



New Virtual Machine Wizard

Select a Guest Operating System

Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows

☒ Linux

☐ VMware ESX

☐ Other

Version

Ubuntu 64-bit

Help

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Next >

Cancel

New Virtual Machine Wizard

Name the Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name:

Ubuntu 64-bit (2)

Location:

C:\Users\suman\OneDrive\Documents\Virtual Machines\Ubuntu 6

Browse...

The default location can be changed at Edit > Preferences.

< Back

Next >

Cancel

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

Ready to Create Virtual Machine

Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.

The virtual machine will be created with the following settings:

Name:	Ubuntu 64-bit (2)
Location:	C:\Users\suman\OneDrive\Documents\Virtual Machines\U...
Version:	Workstation 17.5 or later
Operating System:	Ubuntu 64-bit
Hard Disk:	15 GB, Split
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	CD/DVD, USB Controller, Sound Card

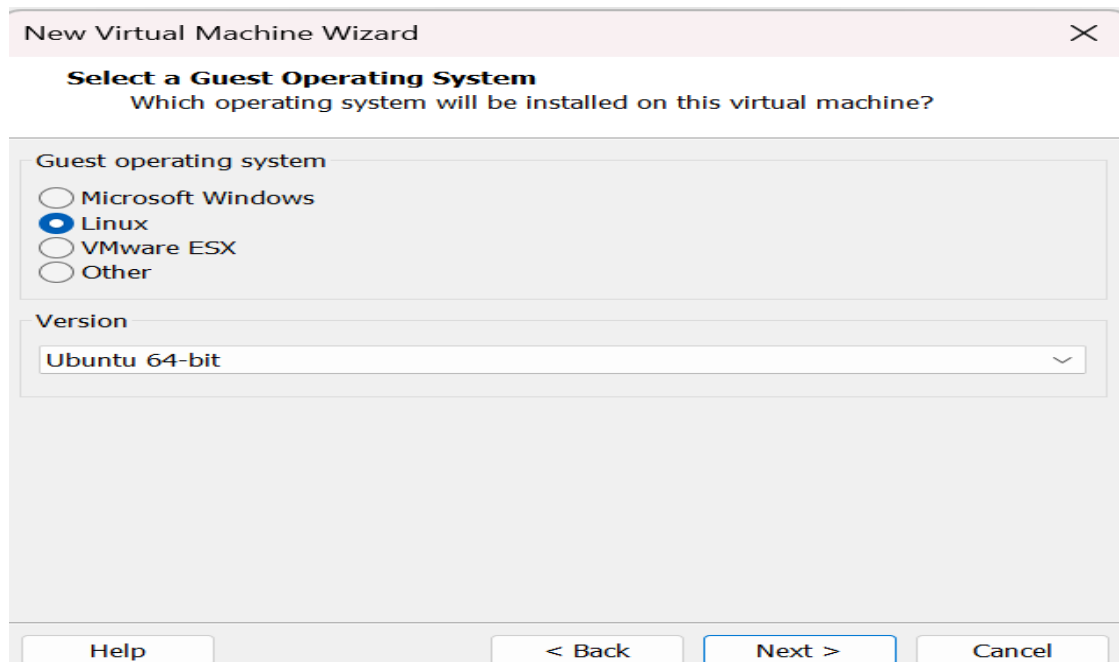
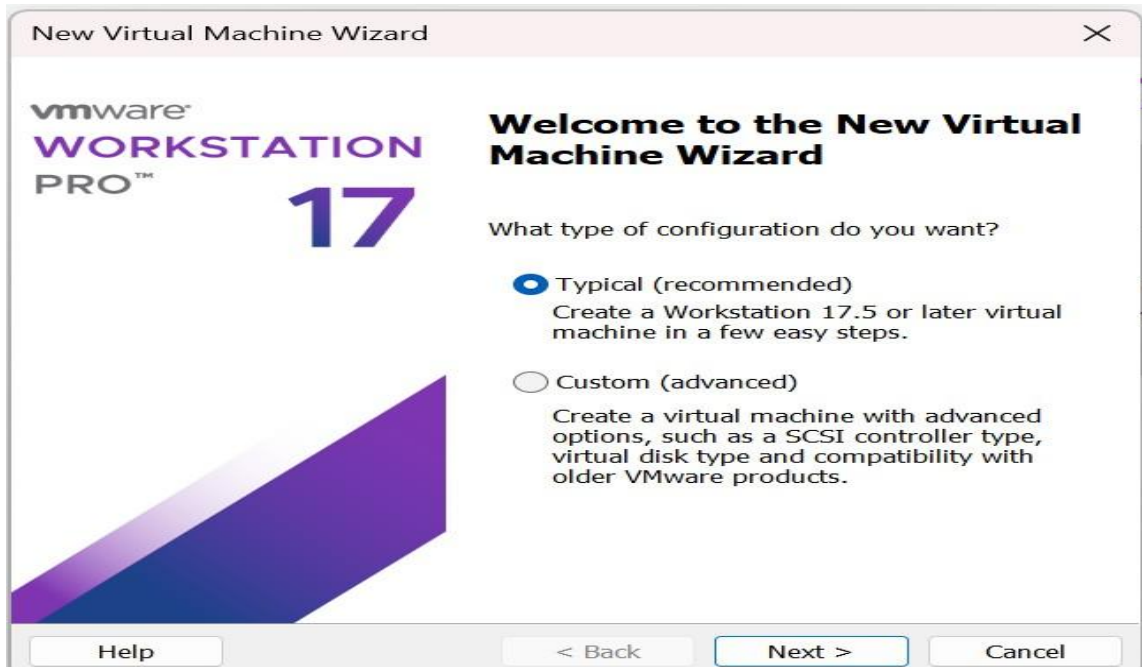
Customize Hardware...

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Finish

Cancel

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor



New Virtual Machine Wizard

×

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB): 

Recommended size for Ubuntu 64-bit: 20 GB

☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

×

Ready to Create Virtual Machine
Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.

The virtual machine will be created with the following settings:

Name:	Ubuntu 64-bit (2)
Location:	C:\Users\suman\OneDrive\Documents\Virtual Machines\U...
Version:	Workstation 17.5 or later
Operating System:	Ubuntu 64-bit
Hard Disk:	15 GB, Split
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	CD/DVD, USB Controller, Sound Card

[Customize Hardware...](#)

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Finish

Cancel

